Towards Computational Fronesis:
Verifying Contextual Appropriateness of Emotions

Michal Ptaszynski, High-Tech Research Center, Hokkai-Gakuen University, Sapporo, Hokkaidō, Japan

Pawel Dybala, Department of Information and Management Science, Otaru University of Commerce, Sapporo, Hokkaidō, Japan

Michal Mazur, Graduate School of Information Science and Technology, Hokkaido University, Sapporo, Hokkaidō, Japan

Rafal Rzepka, Graduate School of Information Science and Technology, Hokkaido University, Sapporo, Hokkaidō, Japan

Kenji Araki, Graduate School of Information Science and Technology, Hokkaido University, Sapporo, Hokkaidō, Japan

Yoshio Momouchi, Department of Electronics and Information Engineering, Faculty of Engineering, Hokkai-Gakuen University, Sapporo, Hokkaidō, Japan

ABSTRACT

This paper presents research in Contextual Affect Analysis (CAA) for the need of future application in intelligent agents, such as conversational agents or artificial tutors. The authors propose a new term, Computational Fronesis (CF), to embrace the tasks included in CAA applied to development of conversational agents such as artificial tutors. In tutor-student discourse it is crucial that the artificial tutor was able not only to detect user/student emotions, but also to verify toward whom they were directed and whether they were appropriate for the context of the conversation. Therefore, as the first task in CF the authors focus on verification of contextual appropriateness of emotions. They performed some of the first experiments in this task for the Japanese language and discuss future directions in development and implications of Computational Fronesis.

Keywords: *Affect Analysis, Applied E-Learning, Artificial Intelligence, Computational Fronesis (CF), Computational Linguistics, Contextual Affect Analysis (CAA)*

DOI: 10.4018/jdet.2013040102

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INTRODUCTION

In recent years there has been a major development in distant education technologies, such as E-learning, Online learning, Computer-based learning (CBL) or Technology-enhanced learning (TEL). In general E-learning aims at supporting or replacing usual classroom education with the use of modern information and communication technology. It is estimated that the E-learning industry has grown to be worth over 27 billion dollars (Ontario Media Development Corporation, 2010) and is estimated to exceed 100 billion in 2015 (Global Industry Analysts, 2010). It has also gained on popularity among students, who prefer E-learning courses to the usual courses in 25% of cases (Allen & Seaman, 2008).

One kind of E-learning is a virtual classroom lead either by a human teacher or, most recently, by an artificial tutor. These can be realized in Virtual Learning Environments (VLE), created on the basis of chat-rooms, either through text-based communication, or with the support of voice and online presentations. There have been also attempts to use virtual worlds, such as Second Life as a base for VLE (Redecker, 2009).

In such virtual classrooms a great role is assigned to knowledge acquisition through social interaction. Contrary to usual classroom scenario, where a teacher stands before students, gives a lecture and asks the students questions, VLE offer a kind of learning that is more interactive and emotionally engaging for students. Learning through conversation (Laurillard, 1993; Laurillard, 1999) with regards to students' emotional engagement (Baath, 1982) are thus two important issues in E-learning. These two issues have lead us in this research. We focused on supporting conversational artificial tutors with knowledge on user emotions. In particular we developed a method for context aware emotion recognition to be applied in such tutors. The method is responsible to verify whether emotions expressed by human users are appropriate to the context of conversation.

The outline of this paper is as follows. First we present the background for this research. We then describe the fields of Artificial Tutors (AT) development, and Contextual Affect Analysis (CAA). Afterwards, we also propose a novel term "Computational Fronesis" embracing the tasks of supporting AT with CAA. Followed by a section in which we describe previously developed tools applied in the research. Next we describe the method for verifying whether the emotions expressed in conversations are appropriate to the context of the situation. The two sections after contain descriptions of the design of evaluation experiment and its results, respectively. Then we present a discussion on further implication of context aware Affect Analysis. Finally, we conclude the paper and propose some ideas to improve the described method and point out some of the possible future applications.

BACKGROUND

Artificial Tutors

The research on artificial tutors is a relatively new field. Several artificial tutors have been developed in the form of non-task oriented conversational agents for second language acquisition (Jia, 2009); Tatai, Csordas, Kiss, Szalo, & Laufer, 2003; Stewart & File, 2007). In such research the pedagogical conversational agent/tutor assists students and provides a support to their learning efforts. In particular, Fryer et al. recently assessed the usefulness of chatterbots as language learning tools (Fryer & Carpenter, 2006).

There have been several VLE incorporating conversational agents to pedagogical tasks. CSIEC (Computer Simulation in Educational Communication) is an Artificial Intelligence (AI) framework developed at Peking University (Jia, 2009). It provides learners with a chatting partner for educational purposes. However, it deals with some problems regarding the ability to understand and generate natural language

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caused by textual ambiguity. *Let's Chat* is a prototype of a computer dialogue system allowing students to practice second language in social situations (Stewart & File, 2007). However, it uses only pre-stored utterances, so conversation topics in the dialogue system are significantly limited. A conversational language tutor in the form of an AI chatterbot CLIVE was presented by Zakos et al. (2008). CLIVE accepts input messages in two languages and thereby allows learning also for students with a limited knowledge of the second language.

Recently it has been noticed that equipping artificial tutors with knowledge on user/student emotions improves the overall impression of artificial tutors. Especially Florea (1999) and Florea and Kalisz (2005) proposed an artificial tutor integrated in an e-learning environment. The tutor develops lecture plans and tailors its behavior according to student's affective states. Vincente and Pain (2002) showed later that the recognition of emotions applied in artificial tutors can be useful in predicting students' motivation.

However, to guide the students appropriately, an artificial tutor, needs to have more than the ability to recognize emotions. For example, it needs to be able to detect toward whom an emotion is directed. The tutor also needs to have an ability to discriminate whether the expressed emotion is appropriate for the moment, or the context of conversation. In this paper we address this particular issue with regards to Japanese language speaking conversational agents.

Contextual Affect Analysis

Text based Affect Analysis (AA) has been defined as a field focused on developing natural language processing techniques for estimating the emotive aspect of text (Grefenstette, Qu, Shanahan, & Evans, 2004). For example, Elliott (1992) proposed a keyword-based Affect Analysis system applying an affect lexicon (including words like "happy", or "sad") with intensity modifiers (words like "extremely", "somewhat"). Liu et al. (2003) presented a model of text-based affect sensing based on OMCS (Open-Mind Common Sense), a generic common sense database, with an application to e-mail interpretation. Alm et al. (2005) proposed a machine learning method for affect analysis of fairy tales. Aman and Szpakowicz also applied machine learning techniques to analyze emotions expressed on blogs (Aman & Szpakowicz, 2007).

There have also been several attempts to achieve this goal for the Japanese language. For example, Tsuchiya et al. (2007) tried to estimate emotive aspect of utterances with a use of an association mechanism. On the other hand, Tokuhisa et al. (2008) and Shi et al. (2008) used a large number of examples gathered from the Web to estimate user emotions. Furthermore, Ptaszynski et al. (2009a) proposed a Web-based supported affect analysis system for Japanese text-based utterances.

Research on emotions in the fields of Artificial Intelligence and Natural Language Processing, like the ones described above has flourished rapidly through several years. Unfortunately, in much of such research contextuality of emotions is disregarded. Based only on behavioral approaches, methods for emotion recognition ignore the context of emotional expression. Therefore, although achieving good results in laboratory conditions, such methods are often inapplicable in real world tasks. For example, a system for recognition of emotions from facial expressions, assigning "sadness" when user is crying would be critically mistaken if the user was, e.g., cutting an onion in the kitchen. Similarly in language, not including con text in the processing could lead to various processing errors. For example, one can consider a system detecting happiness when user uses a word "happy". The system would be critically mistaken if the user actually said: "I'm not happy at all!". This shows that not considering at least grammatical context in the processing causes erroneous detection of opposite emotion. However, a deeper problem appears when the user said something like "I would be so happy if that bastard was hit by a car!". Here a grammatical context does not

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suffice correct processing and deeper context is required in the processing. As the above examples show, recognizing emotions without recognizing their context is incomplete and cannot be sufficient for real-world applications.

Processing the context of emotions, or Contextual Affect Analysis (CAA) (Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2009b; Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2009c; Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2009d; Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2010; Zhang, 2010a; Zhang, 2010b; Zhang, 2011a; Zhang, 2011b; Zhang, 2011), is a newly recognized field. During its fifteen years of history, research on computer processing of emotions, or Affective Computing (Picard, 1997), was in great part focused on the recognition of expressions of user emotions. However, little research addressed the need for computing the context of the expressed emotions. In the age of information explosion, with an easy access to very large sources of data (such as the Internet), the time has come to finally address this burning need. Our research is focused on only one type of emotion processing, affect analysis of text. The future challenge will be to develop methods for processing the context in more general meaning, making the machines aware of the sophisticated environment humans live in. It has been shown that CAA is a feasible task, although much research in this matter needs to be done in the near future. In this paper, we focused in particular on applying context processing to text-based affect analysis. We did this in two ways.

Firstly, one of the common problems in the keyword-based systems for affect analysis is confusing the valence of emotion types, since the emotive expression keywords are extracted without their grammatical context. An idea aiming to solve this problem is the idea of Contextual Valence Shifters (CVS), words and phrases like "not", or "never", which change the valence of an emotion (positive/negative). As the first step towards contextual processing of emotions we applied CVS as a supporting procedure for affect analysis system for Japanese.

Secondly, we have developed a method making use of the wider context an emotion is expressed in. The method, using a Web mining technique, determines, whether the expressed emotion is appropriate for its context. It introduces an idea of Contextual Appropriateness of Emotions to the research on emotion processing. This idea adds a new dimension in emotion recognition, since it assumes that both positive and negative emotions can be appropriate, or inappropriate, depending on their contexts. The method is based on the assumption that the Internet can be considered as a database of experiences people describe on their homepages or weblogs. Since the context of emotions is formulated through collecting experiences, these experiences could be as well "borrowed" from the Internet (Rzepka & Araki, 2007).

In conclusions to this paper we present a discussion on future directions and applications of context processing within Affective Computing. We also propose a term, Computational Fronesis, to embrace the tasks related to Contextual Affect Analysis with regards to conversational artificial agents-companions, and artificial tutors in particular.

Computational Fronesis

Fronesis (often also spelled as *phronesis* or *phronēsis*), from Greek □ρόνησιζ is a word originally used in Ancient Greek in the meaning of "practical wisdom" or "practical judgment" (sometimes also translated as "prudence"). The concept of fronesis is explained by Aristotle in *Nicomachean Ethics* (Aristotle, 1999). Aristotle defines it as practical judgment used to make good decisions according to a situation or context. He distinguishes *fronesis* from *techne*, which is a specific/expert knowledge and *sophia* ("theoretical wisdom"). Where *sophia* can be acquired by pure study/learning, and *techne* can be acquired by practice of some specific tasks, *fronesis* is a knowledge about everyday matters that can be acquired only by experience. This includes behaving appropriately to the occasion, although not always in a self-controlled

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and patient way. Rather the reaction, whether it is calm or explosive, respectful and modest or contemptuous and looking down on somebody, should be chosen according to the other person's behavior, their status and the overall situation. Solomon (1993) in his treaty on emotions and intelligence gives a number of examples for *fronesis*. For example, one could get angry at someone for making a fool of him in front of other people. However, if the person has significantly higher status (e.g., one's boss or supervisor), expressing the anger could end in losing a job. In such situation getting angry instead at ones children or pets is inappropriate. On the other hand, if the insult was said by a person of equal status (a colleague), the anger should be expressed reasonably. However, if the insulting person is of lower status (a subordinate), it is appropriate not only to react with great anger, but the anger should be expressed together with contempt and a lesson should be given to the person breaking the social rules. Moreover, if one does not react in this appropriate way, one could even be perceived as a fool (a person lacking *fronesis*). Therefore *fronesis* with regards to emotions is a kind of "good judgment", including knowledge on how to express ones emotions, towards whom they ought to be expressed, in which situations, and when it is appropriate to express them.

When it comes to artificial tutors interacting with human users, or conversational agents in general, it is often argued that these should be aware of their social role. Therefore, we find proposing a computational model of fronesis not only important, but crucial to the task of creating a fully socialized artificial tutor with human-like behavior. Therefore we propose a term "Computational Fronesis" (CF) as a general notion to deal with all kinds of contextual information in the research on Affect Analysis and computer processing of emotions in general. In particular, we propose an initial set of tasks in form of research questions below[1]:

1. Who expresses the emotion?
2. Why they express the emotion?
3. Is the expression of emotion appropriate to the situation/context?
4. Is the degree of expression appropriate to the situation/context?
5. If the expression is not appropriate, what would be the appropriate one?

These initial five questions can be put in a form of abilities or tasks, such as the ones below, respectively:

1. Determination of emotion subject;
2. Determination of emotion object;
3. Verification of contextual appropriateness of emotions;
4. Verification of appropriateness of the degree/intensity of the expressed emotion;
5. Emotion-related behavioral pattern modeling.

In this paper we focus on task III..We address task I. (Dokoshi, Oyama, Kurihara, Ptaszynski, Rzepka, & Araki, 2011) and II. (Ptaszynski, Dybala, Rzepka, Araki, & Momouchi, 2012; Ptaszynski, Rzepka, Araki, & Momouchi, 2012) in a different research. In the future we plan to address the two other tasks (verification of appropriateness of emotion intensity and behavioral pattern modeling) as well.

AFFECT ANALYSIS TOOLS

In this section we describe all tools and methods for basic AA used further in CAA tasks.

Emotive Expression Dictionary

Emotive Expression Dictionary (Nakamura, 1993) is a dictionary developed by Akira Nakamura in a period of over 20-year time. It is a collection of over two thousand expressions describing emotional states collected manually from a wide range of literature. It is not a tool per se, but was converted into an emotive expression database by Ptaszynski et al. (2009a; 2009) in their research on affect analysis of utterances

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in Japanese. In our task we needed to choose a lexicon proved to be the most appropriate for the Japanese language. Nakamura's dictionary is a state-of-the art example of a handcrafted lexicon of emotive expressions. It also proposes a classification of emotions that reflects the Japanese language and culture the most appropriately. In particular, Nakamura proposes ten emotion types: 喜 *ki/yorokobi*[2] (joy, delight; later referred to as *joy*), 怒 *dō/ikari* (*anger*), 哀 *ai/aware* (sorrow, sadness, gloom; later referred to as *sadness*), 怖 *fu/kowagari* (*fear*), 恥 *chi/haji* (shame, shyness, bashfulness; later referred to as *shame*), 好 *kō/suki* (liking, fondness; later referred to as *fondness*), 厭 *en/iya* (dislike, detestation; later referred to as *dislike*), 昂 *kō/takaburi* (*excitement*), 安 *an/yasuragi* (*relief*), and 驚 *kyō/odoroki* (surprise, amazement; later referred to as *surprize*). This classification is also applied in the lexicon itself. All expressions are classified as representing a specific emotion type, one or more if applicable. The distribution of separate expressions across all emotion classes is represented in Table 1.

A frequent manner in text-based Affect Analysis research is applying a list of emotion classes based on other modalities than linguistic, such as face recognition, or simply creating a new class list for the need of a particular research (see for example comparison of emotion class standards in Ptaszynski, Dybala, Rzepka, Araki, and Momouchi (2012)). In our research we aimed at contributing to the standardization of emotion class list in the language based research on emotions in Japanese. Therefore we selected Nakamura's emotion type list, developed for over 20 years, as the most appropriate for our research.

ML-Ask

ML-Ask, or *eMotive eLement and expression **A**nalysis **s**ystem* is a keyword-based language-dependent system for automatic affect annotation on utterances in Japanese constructed by Ptaszynski et al. [24, 42]. It uses a two-step procedure:

1. Specifying whether an utterance is emotive; and
2. Recognizing the particular emotion types in utterances described as emotive.

ML-Ask is based on the idea of two-part classification of realizations of emotions in language into:

1. *Emotive elements* or *emotemes*, which indicate that a sentence is emotive, but do not detail what specific emotions have been expressed. For example, interjections such as "whoa!" or "Oh!" indicate that the speaker (producer of the utterance) have conveyed some emotions. However, it is not possible, basing only on the analysis of those words, to estimate precisely what kind of emotion the speaker conveyed. Ptaszynski et al. (2009) include in emotemes such groups as interjections, mimetic

Table 1. Distribution of emotive expressions across emotion classes in Nakamura's dictionary, ordered by the number of expressions per class

Emotion Class	Nunber of Expressions	Emotion Class	Nunber of Expressions
dislike	532	fondness	197
excitement	269	fear	147
sadness	232	surprise	129
joy	224	relief	106
anger	199	shame	65
		sum	**2100**

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expressions, vulgar language and emotive markers. The examples in Japanese are respectively: *sugee* (great! - interjection), *wakuwaku* (heart pounding - mimetic), *-yagaru* (syntactic morpheme used in verb vulgarization) and '!', or '??' (sentence markers indicating emotiveness). Ptaszynski et al. collected and hand-crafted a database of 907 emotemes. A set of features similar to what is defined by Ptaszynski et al. as emotemes has been also applied in other research on discrimination between emotive (emotional/subjective) and non-emotive (neutral/objective) sentences (Wiebe, Wilson, & Cardie, 2005; Aman & Szpakowicz, 2007; Wilson & Wiebe, 2005);

2. Emotive expressions are words or phrases that directly describe emotional states, but could be used to both express one's emotions and describe the emotion without emotional engagement. This group could be realized by such words as *aijou* (love - noun), *kanashimu* (feel sad, grieve - verb), *ureshii* (happy - adjective), or phrases such as: *mushizu ga hashiru* (to give one the creeps [of hate]) or *ashi ga chi ni tsukanai* (walk on air [of happiness]). As the collection of emotive expressions ML-Ask uses a database created on the basis of Nakamura's *Emotive Expression Dictionary* (Nakamura, 1993).

With these settings ML-Ask was proved to distinguish emotive sentences from non-emotive with a very high accuracy (over 90%) and to annotate affective information on utterances with a sufficiently high Precision (85.7% compared to human annotators), and satisfying, although not ideal Recall (54.7%) (Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2009;(Ptaszynski, Dybala, Rzepka, & Araki, 2009). To improve the system performance we also implemented Contextual Valence Shifters.

The idea of Contextual Valence Shifters (CVS) was first proposed by Polanyi and Zaenen (2006). They distinguished two kinds of CVS: negations and intensifiers. The group of negations contains words and phrases like "not", "never", and "not quite", which change the valence (also called polarity or the semantic orientation) of an evaluative word they refer to. The group of intensifiers contains words like "very", "very much", and "deeply", which intensify the semantic orientation of an evaluative word. ML-Ask fully incorporates the negation type of CVS with a 108 syntactic negation structures. Examples of CVS negations in Japanese are structures such as: *amari -nai* (not quite-), *-to wa ienai* (cannot say it is-), or *-te wa ikenai* (cannot [verb]-). In this paper we compared the performance of ML-Ask with and without (baseline) CVS improvement, within the evaluation of the procedure for verification of emotion appropriateness. As for intensifiers, although ML-Ask does not include them as a separate database, most Japanese intensifiers are included in the emoteme database. The system calculates emotive value, which is interpretable as emotional intensity of a sentence. It is calculated as the sum of emotemes in the sentence. The performance of setting the emotive value was evaluated on 84% comparing to human annotators (Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2009). Finally, the last distinguishable feature of ML-Ask is implementation of Russell's two dimensional affect space (Russell, 1980). It assumes that all emotions can be represented in two dimensions: the emotion's valence (positive/negative) and activation (activated/deactivated). An example of negative-activated emotion could be "anger"; a positive-deactivated emotion is, e.g., "relief". The mapping of Nakamura's emotion types on Russell's two dimensions proposed by Ptaszynski et al. (2009) was proved reliable in several research (Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2009b; Ptaszynski, Dybala, Rzepka, & Araki, 2009; Ptaszynski, Maciejewski, Dybala, Rzepka, & Araki, 2010). The mapping is represented in Figure 1. An example of ML-Ask output is represented in Figure 2.

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Figure 1. Mapping of Nakamura's classification of emotions on Russell's 2D space

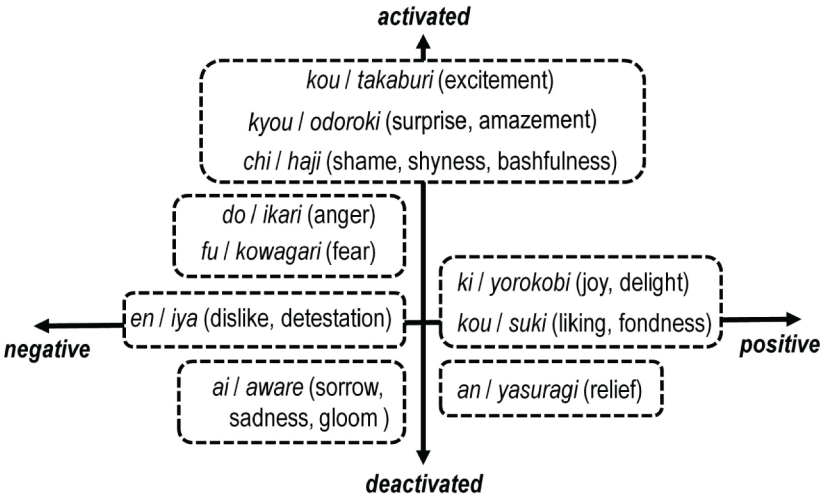


Figure 2. Output examples for ML-Ask and CAO

Sentence: なぜかレディーガガを見ると恐怖感じる(;´艸`)
Spaced: なぜか レディーガガ を 見る と 恐怖 感じる (;´艸`)
Transliteration: *Nazeka Lady Gaga wo miru to kyoufu kanjiru* (;´艸`)
Translation: Somehow Lady Gaga frightens me (;´艸`)

AFFECTIVE INFORMATION ANNOTATIONS

CAO output:
Extracted emoticon: (;´艸`)
Emoticon segmentation:
S1 | BL|S2|ELMER| S3 |BR|S4
N/A| (|;| ´艸` |N/A|)|N/A

Emotion score
Fear (0.02708333)
Surprize (0.01973684)
Dislike (0.0105364)
Excitement (0.01018174)

Anger (0.00703125)
Sorrow (0.004665203)
Shame (0.004424779)
Joy (0.002962932)
Fondness (0.00185117)
Relief (0)

ML-Ask output: なぜかレディーガガを見ると恐怖感じる(;´艸`)
sentence: emotive
emotemes: EMOTICON:(;´艸`)
emotions:(1),FEAR:恐怖
2D: NEGATIVE, ACTIVE

CAO

CAO, or *emotiCon Analysis and decOding of affective information system* is a system for estimation of emotions conveyed through emoticons[3] developed by Ptaszynski et al. (2010). Emoticons are sets of symbols widely used to convey emotions in text-based online communication, such as blogs. CAO extracts an emoticon from an input (a sentence) and determines specific emotion types expressed by it using a three-step procedure. Firstly, it matches the input to a predetermined raw emoticon database containing over ten thousand emoticons. The emoticons, which could not be estimated using only the database are automatically divided into semantic areas, such as representations of "mouth" or "eyes", basing on the idea of *kinemes*, or minimal meaningful body movements, from the theory of kinesics (Birdwhistell, 1952; Birdwhistell, 1970). The areas are automatically annotated according to their co-occurrence in the database. The annotation is firstly based on eye-mouth-eye triplet.

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If no triplet was found, all semantic areas are estimated separately. This provides hints about potential groups of expressed emotions giving the system coverage of over 3 million possibilities. The performance of CAO was evaluated as close to 98% (Ptaszynski, Maciejewski, Dybala, Rzepka, & Araki, 2010) which proved CAO as a reliable tool for the analysis of Japanese emoticons. In the annotation process CAO was used as a supporting procedure in ML-Ask to improve the performance of the affect annotation system and add detailed information about emoticons appearing in the text. An example of CAO output is represented in Figure 2.

Web Mining Technique for Emotion Association Extraction

To verify the appropriateness of the speaker's affective states we applied Shi et al.'s (2008) Web mining technique for extracting emotive associations from the Web. Ptaszynski et al. (2009a) already showed that ML-Ask and Shi's technique are compatible and can be used as complementary means to improve the emotion recognition task. However, these two methods are based on different assumptions. ML-Ask is a language based affect analysis system and can recognize the particular emotion expression conveyed by a user. On the other hand, Shi's technique gathers from the Internet large number of examples and derives from this data an approximated reasoning about what emotion types usually associate with the input contents. Therefore it is more reasonable to use the former system as emotion detector, and the latter one as a verifier of naturalness, or appropriateness of user emotions.

Shi's technique performs common-sense reasoning about which emotions are the most natural to appear in the context of an utterance, or in other words, which emotions should be associated with it. Emotions expressed, which are unnatural for the context (low or not on the list) are perceived as inappropriate. The technique is composed of three steps: 1) extracting context phrases from an utterance; 2) adding causality morphemes to the context phrases; 3) cross-referencing the modified phrases on the Web with emotive lexicon and extracting emotion associations for each context phrase.

Phrase Extraction Procedure

An utterance is first processed by MeCab, a tool for tokenization and part-of-speech analysis of Japanese (Kudo, 2001). Every element separated by MeCab is treated as a unigram. All unigrams are grouped into larger groups of n-grams preserving their word order in the utterance. The groups are arranged from the longest n-gram (the whole sentence) down to all groups of trigrams. N-grams ending with particles are excluded, since they gave too many ambiguous results in pre-test phase. An example of phrase extraction is presented in Table 2.

Table 2. Example of context n-gram phrases separation from an utterance. Grammar shortcuts: SUB = subject particle, GER = gerund, PRF = perfect form.

Original Utterance **English Translation**	*Aa, pasokon ga kowarete shimatta…* Darn, the PC has broken…					
Longest n-Gram	(1) *Aa*	*pasokon*	*ga*	*koware-*	*te*	*shimau*
(Here: Hexagram)	[interjection]	[noun]	[SUB]	[verb]	[GER]	[PRF]
Pentagram	(2) *pasokon □ga □koware □te □shimau*					
Tetragram	(3) *Aa, □pasokon □ga □kowareru*					
Trigrams	(4) *pasokon □ga □kowareru*				(5) *koware □te□ shimau*	

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Morpheme Modification Procedure

On the list of n-gram phrases the ones ending with a verb or an adjective are then modified grammatically with causality morphemes. This is performed in line with linguistic argument that Japanese people tend to convey emotive meaning after causality morphemes (Yamashita, 1999). Shi et al. (2008) independently confirmed this argument experimentally. They distinguished eleven emotively stigmatized morphemes for the Japanese language using statistical analysis of Web contents and performed a cross reference of appearance of the eleven morphemes with the emotive expression database using the Google search engine. This provided the results (hit-rate) showing which of the eleven causality morphemes were the most frequently used to express emotions. For the five most frequent morphemes, the coverage of Web mining procedure still exceeded 90%. Therefore for the Web mining they decided to use those five ones, namely: *-te, -to, -node, -kara* and *-tara* (see Table 3). An example of morpheme modification is presented in Table 4.

Emotion Association Extraction Procedure

In this step the modified n-gram phrases are used as a query in Google search engine and 100 snippets for one morpheme modification per query phrase is extracted. This way a maximum of 500 snippets for each queried phrase is extracted. These are cross-referenced with emotive expression database (see Figure 3). The emotive expressions extracted from the snippets are collected, and the results for every emotion type are sorted in descending order. This way a list of emotions associated with the queried sentence is obtained. It is the approximated emotive commonsense used further as an appropriateness indicator. An example of emotive association extraction is shown in Table 5.

Blog Mining

The baseline of the Web mining method, using Google to search through the whole Web, was gathering a large amount of noise. To solve this problem we made two modifications. Firstly, we

Table 3. Hit-rate results for each of the eleven morphemes with the ones used in the web mining technique in bold font

Morpheme	*-te*	*-node*	*-tara*	*-nara*	*-kotoga*	*-nowa*
Result	41.97%	7.20%	5.94%	1.17%	0.35%	2.30%
Morpheme	*-to*	*-kara*	subtotal	*-ba*	*-noga*	*-kotowa*
Result	31.97%	6.32%	93.40%	3.19%	2.15%	0.30%

Table 4. Examples of n-gram modifications for web mining

Original n-Gram	**pasokon ga koware te shimau**	**/causality morpheme/**
n-gram phrase adjusting (morpheme modification)	*pasokon ga koware te shimat -te*	/ *-te* /
	pasokon ga koware te shimau -to	/ *-to* /
	pasokon ga koware te shimau -node	/ *-node* /
	pasokon ga koware te shimau -kara	/ *-kara* /
	...	...

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Figure 3. Flow chart of the web mining technique

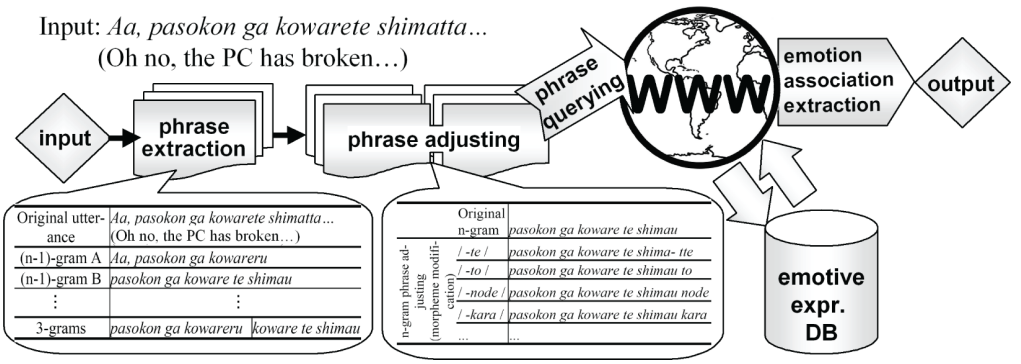


Table 5. Example of emotion association extraction from the web and its improvement by blog mining procedure

Sentence: *Konpyūta wa omoshiroi desu ne.* (Computers are so interesting.)				
Extracted Emotion Types	**Baseline: Type Extracted / All Extracted Types (Ratio)**	**Extracted Emotion Types**	**Blogs: Type Extracted / All Extracted Types (Ratio)**	
Fondness	79/284(0.287)	fondness	601/610(0.985)	
Surprise	30/284(0.105)	excitement	1/610(0.001)	[rejected
Excitement	30/284(0.105)	fear	1/610(0.001)	as
Fear	29/284(0.102)	relief	1/610(0.001)	noise]
...	...	...	...	

added a heuristic rule stopping the search if any emotions were found using the longer n-grams. This changed the method from Recall-oriented to Precision-oriented by assuring the extraction of only the closest emotive associations. It also speeds up the extraction process. Secondly, since, as mentioned before, people convey on blogs their opinions and emotions, we restricted the mining to blog contents to assure extraction of more accurate emotive associations. The blog mining procedure performs the query first on the public blogs from *Yahoo!Japan-Blogs*[4]. The paragraphs of each blog containing query phrases are co-referenced with emotive expression database to gather the emotive associations. If no information was gathered from the blog contents, the same search is performed with the baseline conditions - on the whole Web. An example of improvement is presented in Table 5.

METHOD FOR VERIFICATION OF CONTEXTUAL APPROPRIATENESS OF EMOTIONS

As one of the recent advances in affect analysis, it was shown that Web mining methods can improve the performance of language-based affect analysis systems (Tokuhisa, Inui, & Matsumoto, 2008; Shi, Rzepka, & Araki, 2008; Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2008). However, in these methods, although the results of experiments appear to be positive, two extremely different approaches are mixed, the language/keyword based approach and the Web mining based approach. The former, in which the information provided by the user in input is matched to the existing lexicons and sets of rules, is responsible for recognizing the

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particular emotion expression conveyed by the user at a certain time. The latter one is based on gathering from the Internet large numbers of examples and derives from these an approximated reasoning about what emotions usually associate with certain contents. Using the Web simply as a complementary mean for the language based approach, although achieving reasonable results, does not fully exploit the potential lying in the Web (Rzepka & Araki, 2007).

Here we present a method utilizing these two approaches in a more effective way. The method is capable to analyze affect with regard to a context and estimate whether an emotion conveyed in a conversation is appropriate for the particular situation. In the method we used previously developed systems for affect analysis (ML-Ask and CAO described in the previous section). Next, we used a method for gathering emotive associations from the Web developed by Shi et al. (2008).

Furthermore, we checked several versions of the method to optimize its procedures. Firstly, we checked two versions of ML-Ask, with and without Contextual Valence Shifters. Secondly, we checked two versions of the Web mining technique, one performing the search on the whole Internet and the second one searching only through blogs.

Method Description

Affect Analysis

As the first step of the method for verification of contextual appropriateness of emotions, we used the two affect analysis systems described previously (ML-Ask and CAO). The affect analysis provides information on whether an utterance was emotive or not, and what type of emotion was expressed in the utterance. In a conversation between a user and an agent, the affect analysis is performed on each user utterance in user-agent conversation. For every emotive utterance with specified emotion type a Web mining technique is used as a verifier of emotion appropriateness.

Web Mining

In the second step a list of emotive associations is obtained from the Web. This is done with the use of the Web mining technique described previously. The list contains emotion types that associate with the sentence contents. The emotion types that correlate the most strongly appear on the top of the list. Emotion types with weaker correlations appear lower on the list. Emotion types that do not appear on the list at all are considered as the ones with no correlation with the sentence contents. As the rule of thumb we assumed that the emotion types appropriate for the sentence contents (or context in general) should appear in approximated 50% of all results. An example of such a list is represented in Table 5. It shows that when the Web mining is based on the whole Web, emotions considered as appropriate include fondness, and two more (surprise and excitement). However, when the Web mining is limited to blogs the emotion type extraction is more precise and the result (only fondness) is more accurate. Context phrases with less but frequent emotion types extracted are considered as more straightforward and unambiguous. Context phrases with numerous but less frequent emotion types are considered as more ambiguous.

Assessment and Verification of Contextual Appropriateness of Emotion

The final step is to use the two kinds of information (affect analysis and Web mining) in CAEV[5] procedure for assessing and verifying contextual appropriateness of the expressed emotions. The information obtained by affect analysis systems and the Web mining technique described above is combined as follows. When ML-Ask discovers an emotive utterance and the emotion types are successfully specified, the Web mining technique begins the process of verification of whether the expressed emotions are appropriate to be used in such context. The emotions commonly associated with the

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context of the utterance are extracted from the Web and the emotions reported by ML-Ask from the user utterances are compared to the Web mining results. If an emotion type detected by ML-Ask appears on the list of emotions frequently associated with the context (approx. 50% of the whole extracted list), the emotion expressed by the user is determined to be appropriate for the context. In such situations, a conversational agent equipped with this system could choose a dialog strategy that *sympathizes* with the user. Two hypothetical examples are presented below:

- Positive-appropriate emotion:
  - **User:** I'm so happy I passed the exam!
  - **ML-Ask:** joy; Web mining: joy, excitement
  - **Agent:** Yeah! That's great!
- Negative-appropriate emotion:
  - **User:** I hate him for making a fool of me in front of everyone.
  - **ML-Ask:** dislike; Web mining: anger, dislike
  - **Agent:** You have a reason to be angry.

However, if the verification procedure indicates that the expressed emotion is inappropriate for the context, the agent could undertake different measures, e.g., helping user manage his/her emotions, for example by changing the focus of the conversation from the object of emotion to the expressed emotion itself, or proposing an appropriate emotion. Two hypothetical examples are shown below:

- Positive-inappropriate emotion:
  - **User:** I'd be happy if that bastard was hit by a car!
  - **ML-Ask:** joy; Web mining: fear, sadness
  - **Agent:** Are you sure that is what you are really feeling?
- Negative-inappropriate emotion:
  - **User:** I won the prize, but I feel so bored.
  - **ML-Ask:** dislike, depression; Web mining: excitement, joy
  - **Agent:** You should be happy!

Two-Dimensional Model of Affect in CAEV Procedure

According to Solomon (1993), people sometimes misunderstand the specific emotion types, but they rarely misunderstand their generally perceived valence. One could, e.g., confuse anger with irritation, but not admiration with detestation. Therefore, we checked if at least the general features matched even when specific emotion types did not match perfectly with the emotive associations. By general features we refer to those proposed by Russell (1980) in the theory of the two-dimensional model of affect (valence and activation). Using the mapping of Nakamura's emotion types on Russell's model we checked whether the emotion types tagged by ML-Ask and CAO belonged to the same space, even if they did not perfectly match the emotive associations gathered from the Web.

EVALUATION EXPERIMENT

Initial Setting

To test the method, we performed an evaluation experiment on two non-task-oriented conversational agents. The first agent is a simple conversational agent which generates responses by 1) using Web-mining to gather associations to the content of user utterance; 2) making propositions by inputting the associations to the prepared templates; and 3) adding modality to the basic propositions to make the utterance more natural. The second agent, based on the first one, generates a humorous response to user utterance every third turn. The humorous response is a pun created by using user input as a seed to gather pun candidates from the Web and inputting the most frequent pun candidates into pun templates (for more detailed description of the agents see the sec-

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tions below and the references). The choice of the agents was deliberate. They differed only in one feature - the humorous responses in the latter one. Humor processing is considered to be one of the most creative human activity and therefore difficult task in Artificial Intelligence (Boden, 1998). Therefore if appropriateness verification is done correctly, it should be easier to perform on the non-humor-equipped agent.

There were 13 participants in the experiment, 11 males and 2 females. All of them were university undergraduate students. The users were asked to perform a 10-turn conversation with both agents. No topic restrictions were made, so that the conversation could be as free and human-like as possible.

Two Conversational Agents: Short Description

Modalin is a non-task-oriented text-based conversational agent for Japanese. It automatically extracts from the Web sets of words related to a conversation topic set freely by a user in his utterance. The association words retrieved from the Web (with accuracy of over 80%) are then sorted by their co-occurrence on the Web, and the most frequent ones are selected to be used further in output generation. In the response generation, the extracted associations are put into one of the pre-prepared response templates. The choice of the template is random, but the agent keeps in its memory the last choice in order not to generate two similar sentence patterns in a row. Finally, the agent adds a modality pattern to the sentence and verifies its semantic reliability. The modality is added from a set of over 800 patterns extracted from chat-room logs. The naturalness of the final form of the response is then verified on the Web with a hit-rate threshold set arbitrary for 100 hits. The agent was developed by Higuchi and colleagues. For further details see Higuchi, Rzepka, and Araki (2008).

Pundalin is a non-task-oriented conversational agent for Japanese, created on the base of Modalin and combined with Dybala's Pun generating system PUNDA (Dybala, Ptaszynski, Higuchi, Rzepka, & Araki, 2008). The system works as follows. From the user's utterance, a base word is extracted and transformed using Japanese phonetic pun generation patterns, to create a phonetic candidate list. The candidate with the highest hit-rate in the Japanese search engine Goo[6] is chosen as the most common word that sounds similar to the base word. Next, the base word and the candidate are integrated into a sentence. The integration is done in two steps, one for each part of the sentence including the base word and the pun candidate, respectively. Firstly, the base phrase is put into one of several pre-prepared templates making up the first half of the sentence. The second half of the sentence is extracted from KWIC on WEB - on-line Keyword-in-context sentence database (Yoshihira, Takeda, & Sekine, 2004) as the shortest latter half of an emotive sentence including the candidate. Every third turn of the conversation, Modalin's output was replaced by a joke-including sentence, generated by the pun generator. Pundalin therefore is a humor-equipped conversational agent using puns to enhance communication with the user. Pundalin was developed by Dybala and colleagues as a conversational agent for use in experiments on the influence of humor on human-agent interaction (Dybala, Ptaszynski, Higuchi, Rzepka, & Araki, 2008).

Overview of Conversations

As mentioned on the beginning of above sections, thirteen participants (11 males and 2 females) performed conversations with the agents. Each participant performed a 10 turn conversation with each of the two agents. Therefore each conversation consisted of 20 utterances. The order of which agent was the first to talk with the user was random. Since we assumed no topic restrictions, the conversations were usually small talks about weather, food or other casual topics. In the process of affect analysis only the utterances performed by human users were analyzed. In comparison with Pundalin, there were different emotional reactions to the presence of humorous responses. Some users

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actively engaged in the pun-generating game with the agent, which resulted in responses of positive emotional load. On the other hand, the users who did not find the jokes interesting tended to respond negatively. The emotion types that were expressed the most often across all conversations were: fondness, joy and dislike. Other emotions that appeared sporadically were sadness and fear. Positive emotions appeared more frequently in conversations with Pundalin than with Modalin. An example of conversation with one agent (Pundalin) is represented in Figure 4.

Experiment Design

In the experiment we used the chat logs of users with Modalin and Pundalin. All 26 conversations were analyzed by ML-Ask and CAO. Six out of all 26 conversations contained no specified emotional states and were excluded from further evaluation process. For the remaining conversations, among which 10 were with Modalin and 10 with Pundalin, the Web mining procedure was carried out to determine whether the emotions expressed by the user were contextually appropriate. We compared four versions of the method: 1) ML-Ask and Web mining baseline; 2) ML-Ask supported with CVS, Web mining baseline; 3) ML-Ask baseline and blog mining; 4) both improvements (affect analysis supported with CVS and blog mining). The difference in results appeared in 5 conversation sets (from the evaluated 20 sets). Then a questionnaire was designed to evaluate how close the results were to human thinking. One questionnaire set consisted of one conversation record and questions inquiring what were: 1) the valence (answer to the question: "Were the emotions expressed in this sentence positive or negative?" [choice of three: POS/NEG/Don't know]) and 2) the specific type of emotions conveyed in the conversation (answer to the question: "What was the specific emotion expressed in this sentence?" [free choice]), and 3) whether they were contextually appropriate (answer to the question: "Were the emotions

Figure 4. An example of conversation with one agent (Pundalin). Analyzed sentence underlined. Extracted emotive expression in bold type font. Grammatical errors translated to reflect errors in original conversation. Puns translated to represent puns in English as well.

Speaker	Utterance in Japanese	English Translation
User	*Iwate no jishin sugokatta mitai desu ne?*	The earthquake in Iwate was shocking, wasn't it?
Agent	*Iya, jishin ga sugoi desu kara*	Oh, earthquakes are shocking, you know
User	*Yamakuzure mo okite taihen sōdesu yo ne?*	It looks like there were landslides too, horrible isn't it?
Agent	*Iya, sore wa iyashii na*	Oh, its so mean
User	*Sodesu ka? Dono atari ga iyashii n desu ka?*	Really? Why do you think its mean?
Agent	*Kureyon wo ore ni katte kure yo n*	Don't cry on me, buy crayon for me
User	*Myakuraku ga nasasugi desu.*	That was just too silly.
Agent	*Sore wa omoshiroi tte*	But it was so funny
User	*Sore hodo* ***omoshiroku mo nakatta****desu yo.*	Well, it **wasn't** that **funny**.
Agent	*Kitai ga tsuyoidesu ka*	Maybe you had too strong expectations?
User	*Sore hodo tsuyoi sorehodo okina kitai wa motte inakatta n desu ga.*	I didn't have that strong that big expectations.
Agent	*Atama ga attamaru*	The head is heating up
User	*Tashika ni atama wa tsukau to omoimasu.*	Yes, I'm sure you use your head.
Agent	*Sore wa isogashii daro*	You must be busy
User	*Sorehodo demo nai desu yo.*	Not that much really.
Agent	*Sore wa takai to ka*	Is it high?
User	*Sonna koto nanno hanashi desu ka?*	What are you talking about?
Agent	*Idō ga ōi do*	It's your lot to move a lot!
User	*Yappari myakuraku naidesu.*	Yeah, you are silly.

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appropriate for the context of the conversation? [choice of three: YES/NO/Don't know] If not, which emotion would be appropriate?" [free choice]). Every questionnaire set was filled by 10 people (undergraduate students, but different from the users who performed the conversations with the agents). This way we obtained 200 conversation sets as gold standard. The five conversation sets, where differences in results appeared for the different versions of the method, were compared to the gold standard separately. Therefore there were 20 questionnaire sets for the baseline method and additional 5 for the conversation sets in which results changed after improvements. With every questionnaire set filled by 10 human evaluators there were a total number of 250 different sets of results evaluated by different people.

For every conversation set we calculated how many of the human evaluators confirmed the system's results. The evaluated items were: A) specific emotion types determination; and B) general valence determination accuracies of affect analysis systems; and the accuracy of the method as a whole (affect analysis verified by Web mining) to determine the contextual appropriateness of C) specific emotion types and D) valence.

Evaluation Criteria

A common problem in emotion processing research is the number of evaluators employed in the evaluation process. In a third person evaluation, it is desirable to engage in the evaluation process as many evaluators as possible to get a wide view on the results, calculate an overall agreement and rectify potential errors. Unfortunately there has been no standard for a desirable number of evaluators. Often the evaluation is limited to, e.g., five people (Tsuchiya, Yoshimura, Watabe, & Kawaoka, 2007), three people (Endo, Saito, & Yamamoto, 2006) or even one (Tokuhisa, Inui, & Matsumoto, 2008). The evaluation with only one person, like in (Tokuhisa, Inui, & Matsumoto, 2008) assumes that if at least one person agrees with the system, the system has performed the evaluated task on a human level. The evaluation criteria where three people are employed (Endo, Saito, & Yamamoto, 2006) usually assume that at least two people from the group of three must agree about the evaluated object. The problem becomes complicated with a larger number of evaluators. In the evaluation performed by Tsuchiya et al. (2007) there were five people employed in the evaluation. According to their explanations, a) if four or five people agreed with the system, the results were positive; b) if three or two people agreed with the system, the results were acceptable; c) if only one person or no person agreed with the system, the results were negative.

It can be easily noticed, that although the number of evaluators grows with the introduced approach, the general idea is that at least one person needs to agree with the system for the results to be positive (first approach), or the results are negative only in the case when one person or less from a larger group agree with the system (two latter approaches). These somewhat lenient conditions in evaluation of emotion processing-related systems comes from the fact that it is difficult to obtain a perfect agreement between people about emotion-related topics, since the cognition of emotions in people is highly subjective and context related.

Therefore we decided to look on the results from a more analytical point of view. In our research, apart from the 13 people who took part in conversations with the agents, we evaluated every questionnaire set 10 times. Then we checked how many people agreed with each other and with the results given by the system. Since every questionnaire set was evaluated 10 times, a number of agreements for each evaluated item (A-D) in all twenty evaluated cases could be from 0 (nobody agreed with the system) to 10 (all people agreed with the system). When comparing the four versions of the method described above, we assumed the better version of the system is the one which achieved more agreements with a larger number of evaluators. We took into the consideration all types of agreement conditions, from ideal (all

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10 people agreed) to the smallest one (at least one person agreed with the system), similarly to the research described above. To visualize the results we named every level of agreement like in Table 6. Each level of agreement assumed that the results are positive if at least the specified number of evaluators agreed with the system. However, we focused the most on two of the ten agreement conditions. Firstly, since the idea of appropriateness is based on the rule of thumb, I checked how much of the cases could be considered as positive examples if at least half of the people (five) agreed. Secondly, as a more severe condition, I checked how much of the cases could be considered as correct examples if eight and more people (80%) agreed with the system. Moreover, to verify the strength of the agreements we independently calculated Fleiss' multi-rater kappa (Fleiss, 1971) between all human evaluators for all sets of the results.

RESULTS

We analyzed three aspects of the results. Firstly, we focused on evaluation of the affect analysis procedure. Although the two affect analysis systems were evaluated separately previously (Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2009a; Ptaszynski, Dybala, Rzepka, & Araki, 2009; Ptaszynski, Maciejewski, Dybala, Rzepka, & Araki, 2010), there was no overall evaluation. This corresponds to items A) and B) investigated in the questionnaire. Secondly, we summarized the results of the CAEV procedure for all of the results considered together. This corresponds to items C) and D) investigated in the questionnaire. Finally, we analyzed the results separately for the two agents to check whether there were any differences between verifying the emotion appropriateness in a usual conversational agent (Modalin) and the joking agent (Pundalin).

Evaluation of Affect Analysis Procedure

The first part of the evaluation process consisted in evaluation of affect analysis procedure. The results were as follows. For all possible agreements of the system with the evaluators about the 20 evaluation sets (200 possible cases of agreement) baseline version of affect analysis obtained 110 (55%) agreements about determining emotion type and 126 (63%) agreements about determining valence. The strength of agreements in this setting was κ=0.66 and κ=0.68, respectively, which indicates substantial

Table 6. Criteria conditions and naming of the levels of agreements. Conditions of the highest interest are represented in bold type font.

Criteria Conditions	No. of People who had to Agree
ideal	all 10
rigorous	at least 9
grand majority	**at least 8**
fair	at least 7
weak majority	at least 6
medium	**at least 5**
optimistic	at least 4
easy	at least 3
lenient	at least 2
negligible	at least 1
no agreement	0

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agreement for both sets according to Landis and Koch interpretation (Landis & Koch, 1977). As for the affect analysis procedure upgraded with CVS (ML-Ask last part of the procedure), there were 120 agreements (60%) for emotion types and 138 (68%) for valence determination. The strength of these sets of agreements was κ=0.66 and κ=0.68, respectively. As for the distribution of the agreements, most of the results for emotions types (over 50% of all actual agreements) were enclosed in a group where at least 8 people agreed with the system (grand majority conditions). Similarly, most of the results for valence were enclosed in a group where at least 9 people agreed with the system (nearly ideal, rigorous conditions). For the negligible condition ("at least one person"), often applied in other research, the results enclosed 100% of cases. However, ideal conditions (all agree) appeared from 18% to 29% of cases, which shows that the negligible condition is far from objectivity. However, the grand majority of the results (over 80%) were enclosed in a group where at least 6 people agreed with the system. The conditions including medium (at least 5 people) and more relaxed conditions enclosed from nearly 90% and above. The results are represented in Table 7. Visualization of the distribution of agreements for both versions of affect analysis procedure is represented on Figure 5 (for emotion type determination) and Figure 6 (for valence determination).

Evaluation of CAEV Procedure: General

Secondly we checked the results for the determination of emotion appropriateness by the CAEV procedure. The results were as follows. For all possible agreements of the system with

Table 7. Results for evaluation of affect analysis procedure. Upper part of the table: results for specifying emotion types*; Lower part: results for specifying* valence*. The table represents numbers of people who agreed with the system. Distribution of numbers shows how many there were agreements with how many people;* %of all*: shows percentage of this group of agreements within all agreements;* %sums*: shows percentage of results applicable when the condition for agreement was set as "at least this group of agreements (or higher)";* agr.ratio*: overall number of agreements divided by ideal number of agreements and ratio;* kappa*: multi-rater kappa, or inter-rater agreement coefficient, representing the strength of agreements in this setting.*

A)	Number of people who agreed with the system											agr.ratio
TYPES	10	9	8	7	6	5	4	3	2	1	0	(kappa)
baseline	2	0	5	2	3	1	2	1	0	2	2	110/200
% of all	18%	0%	36%	13%	16%	5%	7%	3%	0%	2%	0%	55%
% sums	18%	18%	55%	67%	84%	88%	95%	98%	98%	100%	100%	(κ= 0.66)
cvs	3	0	5	2	3	1	2	1	0	2	1	120/200
% of all	25%	0%	33%	12%	15%	4%	7%	3%	0%	2%	0%	60%
% sums	25%	25%	58%	70%	85%	89%	96%	98%	98%	100%	100%	(κ= 0.66)

B)	Number of people who agreed with the system											agr.ratio
VALENCE	10	9	8	7	6	5	4	3	2	1	0	(kappa)
baseline	3	4	2	1	2	3	0	2	2	0	1	126/200
% of all	24%	29%	13%	6%	10%	12%	0%	5%	3%	0%	0%	63%
% sums	24%	52%	65%	71%	80%	92%	92%	97%	100%	100%	100%	(κ= 0.68)
cvs	4	4	2	1	2	3	0	2	2	0	0	136/200
% of all	29%	26%	12%	5%	9%	11%	0%	4%	3%	0%	0%	68%
% sums	29%	56%	68%	73%	82%	93%	93%	97%	100%	100%	100%	(κ= 0.68)

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Figure 5. Visualization of the distribution of agreements for both versions of affect analysis procedure in determining about emotion types. Figure corresponds to the upper part of Table 7.

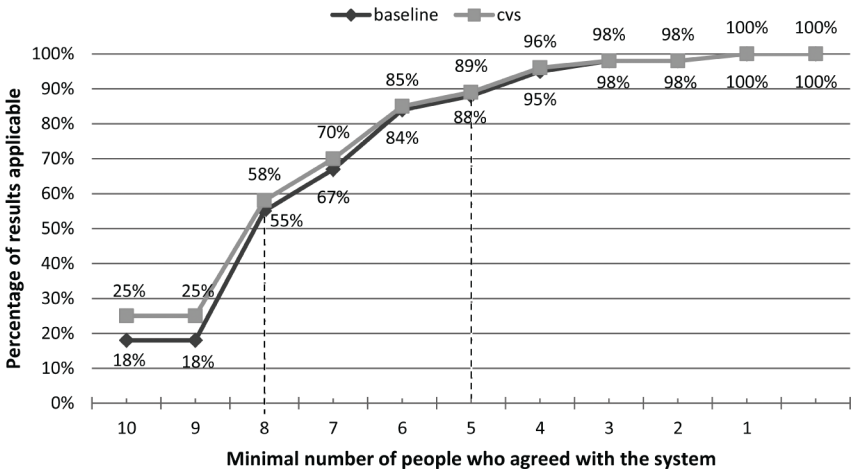
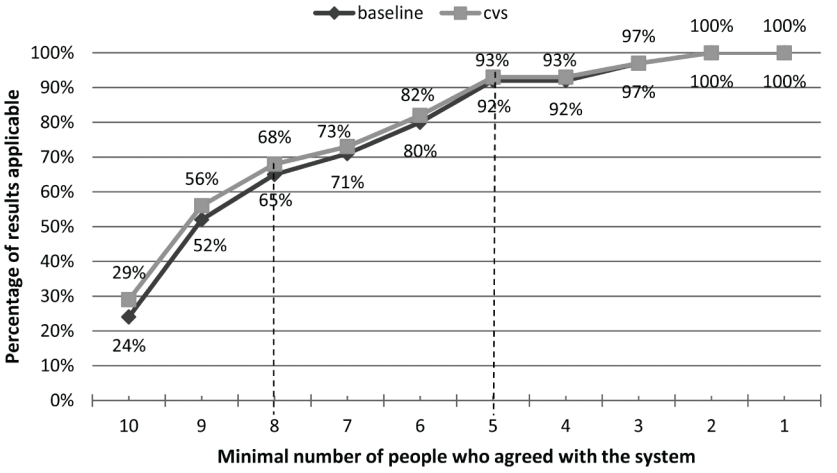


Figure 6. Visualization of the distribution of agreements for both versions of affect analysis procedure in determining about emotion valence. Figure corresponds to the lower part of Table 7.



the evaluators about the 20 evaluation sets (200 possible cases of agreement) baseline version of CAEV procedure obtained 69 (35%) agreements about determining the appropriateness of emotion type and 85 (43%) agreements about determining the appropriateness of valence. The strength of agreements in this setting was κ=0.652 and κ=0.677, respectively, which indicates substantial agreements. As for the version of CAEV procedure with affect analysis upgraded with CVS, there were 78 agreements (39%) for emotion type- and 94 (47%) for valence based appropriateness determination. The strength of these sets of agreements was κ=0.642 and κ=0.667, respectively. As for the version of CAEV procedure with Web mining restricted to blogs, there were 81 agreements (41%) for emotion type and 95 (48%) for valence-based appropriateness determination. The strength of these sets of agreements was κ=0.667 and

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κ=0.643, respectively. Finally, for the version of CAEV procedure with both improvements (ML-Ask upgraded with CVS and Web mining restricted to blogs), there were 90 agreements (45%) for emotion type- and 104 (52%) for valence-based determination. The strength of these sets of agreements was κ=0.643 and κ=0.633, respectively.

As for the distribution of the agreements, the majority of the results (over 50% of all actual agreements) for determining about appropriateness of emotion types were enclosed in a group where at least 5 people agreed with the system (medium conditions). For determining about valence appropriateness, most of the results were enclosed in a group where at least 8 people agreed (grand majority conditions). The results which enclosed at least 80% of agreements oscillated for emotion types verification around groups where at least three (easy) to four (optimistic) people agreed. For valence verification the groups enclosing at least 80% of agreements oscillated from optimistic (at least 4) to medium (at least 5) condition group. Although there were no cases with ideal conditions, the best version of the system (both improvements) encapsulated with the use of grand majority condition (at least 8 people) 48% of results for emotion types and 64% for valence.

The results are represented in Table 8. Visualization of the distribution of agreements for all four versions of CAEV procedure is represented on Figure 8 (for emotion type determination) and Figure 10 (for valence determination). The visualization of percentage of results encapsulated for each condition (from "at least 9" to "at least 1") is presented on Figure 7 (for emotion type determination) and Figure 9 (for valence determination). Some of the successful examples are represented in Table 12.

Evaluation of CAEV Procedure: Agents Separately

Finally, we checked the results for the verification of emotion appropriateness by the CAEV procedure separately for each agent, Modalin and Pundalin. This was done to find out whether verification of emotion appropriateness differs for different types of conversations. Modalin, designed by Higuchi et al. (2008), is a non-task oriented keyword-based conversational agent, which uses modality to enhance dialog propositions extracted from the Web. Apart from this, the agent has no distinctive features. Pundalin is also a non-task oriented conversational agent. It was created by adding to Modalin a pun generating system developed by Dybala et al. (2008). Therefore the only distinctive feature of Pundalin with comparison to Modalin was using puns. We compared the results achieved by the agents separately to check whether the presence of jokes (puns) helps or interrupts the process of verification of emotion appropriateness.

The results were as follows. The overall number of agreements with human evaluators about verification of contextual appropriateness was higher for Modalin (from 41% to 55%) than for Pundalin (from 28% to 49%). This was true for both, emotion types and valence. Also the number of agreements encapsulated for different conditions showed similar tendency. The conditions which encapsulated at least 50% of agreements were, for Pundalin/emotion types, from medium (at least 5 people agreed) to rigorous (at least 9 people agreed). For Pundalin/valence, the results were from medium to grand majority (at least 8 people agreed). The same results for Modalin were approximately higher. For emotion types the condition that encapsulated most of the agreements (over 70%) was a stable medium condition, and for valence it was a stable condition of grand majority (at least 8 people agreed). The conditions that enclosed over 80% of the results oscillated around optimistic and medium conditions. The strength of agreements was considerably high with kappa oscillating from 0.573 to 0.697 across all separate results. The results are summarized for Modalin in Table 9 and for Pundalin in Table 10.

Both improvements, the one with CVS procedure and the one limiting the query scope in the Web mining procedure to search only through blog contents, positively influenced the

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Table 8. Results for evaluation of Contextual Appropriateness of Emotion Verification (CAEV) Procedure. Upper part of the table: results for specifying emotion types*; Lower part: results for specifying* valence*. The Table presents numbers of people who agreed with the system. Distribution of numbers shows how many there were agreements with how many people;* % of all*: shows percentage of this group of agreements within all agreements;* %sums*: shows percentage of results considered when the condition for agreement was set as "at least this group of agreements (or higher)";* agr.ratio*: overall number of agreements divided by ideal number of agreements and ratio;* kappa*: strength of agreements in this setting.*

C) TYPES	Number of people who agreed with the system 9	8	7	6	5	4	3	2	1	0	agr.ratio (kappa)
BASELINE	1	2	0	0	4	2	2	4	2	3	69/200
% of all	13%	23%	0%	0%	29%	12%	9%	12%	3%	0%	35%
% sums	13%	36%	36%	36%	65%	77%	86%	97%	100%	100%	(κ= 0.652)
CVS	2	2	0	0	4	2	2	4	2	2	78/200
% of all	23%	21%	0%	0%	26%	10%	8%	10%	3%	0%	39%
% sums	23%	44%	44%	44%	69%	79%	87%	97%	100%	100%	(κ= 0.642)
BLOGS	2	2	0%	0%	4	3	2	4	1	2	81/200
% of all	22%	20%	0%	0%	25%	15%	7%	10%	1%	0%	41%
% sums	22%	42%	42%	42%	67%	81%	89%	99%	100%	100%	(κ= 0.667)
CVS+BLOGS	3	2	0%	0%	4	3	2	4	1	1	90/200
% of all	30%	18%	0%	0%	22%	13%	7%	9%	1%	0%	45%
% sums	30%	48%	48%	48%	70%	83%	90%	99%	100%	100%	(κ= 0.643)

D) VALENCE	Number of people who agreed with the system 9	8	7	6	5	4	3	2	1	0	agr.ratio (kappa)
BASELINE	2	4	0	0	4	0	4	1	1	4	85/200
% of all	21%	38%	0%	0%	24%	0%	14%	2%	1%	0%	43%
% sums	21%	59%	59%	59%	82%	82%	96%	99%	100%	100%	(κ= 0.677)
CVS	3	4	0	0	4	0	4	1	1	3	94/200
% of all	29%	34%	0%	0%	21%	0%	13%	2%	1%	0%	47%
% sums	29%	63%	63%	63%	84%	84%	97%	99%	100%	100%	(κ= 0.667)
BLOGS	2	5	0	1	2	1	4	2	1	2	95/200
% of all	19%	42%	0%	6%	11%	4%	13%	4%	1%	0%	48%
% sums	19%	61%	61%	67%	78%	82%	95%	99%	100%	100%	(κ= 0.643)
CVS+BLOGS	3	5	0	1	2	1	4	2	1	1	104/200
% of all	26%	38%	0%	6%	10%	4%	12%	4%	1%	0%	52%
% sums	26%	64%	64%	70%	80%	84%	95%	99%	100%	100%	(κ= 0.633)

performance of the Contextual Appropriateness of Emotion Verification procedure, in all of the cases for both of the agents. The improvement was noticeable both on the level of specific emotion types and of valence, and also for the result of both agents taken together as well as separately.

The most effective version of the method was the one with both improvements applied, by which the system's performance (number of agreements with evaluators) was generally improved for all considered cases. For example, for the grand majority condition (at least 8 people agreed) the results were improved from 36% to 48% (emotion types) and from 59% to 64% (valence), with the highest score achieved in conversations with Modalin (75% under the "grand majority" condition).

The results were generally better in Modalin. This confirms the assumption we made on the beginning of the previous section.We assumed that since humor is one of the most

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Figure 7. Visualization of percentage of results encapsulated for each condition, from "at least 9" to "at least 1" (for emotion type determination)

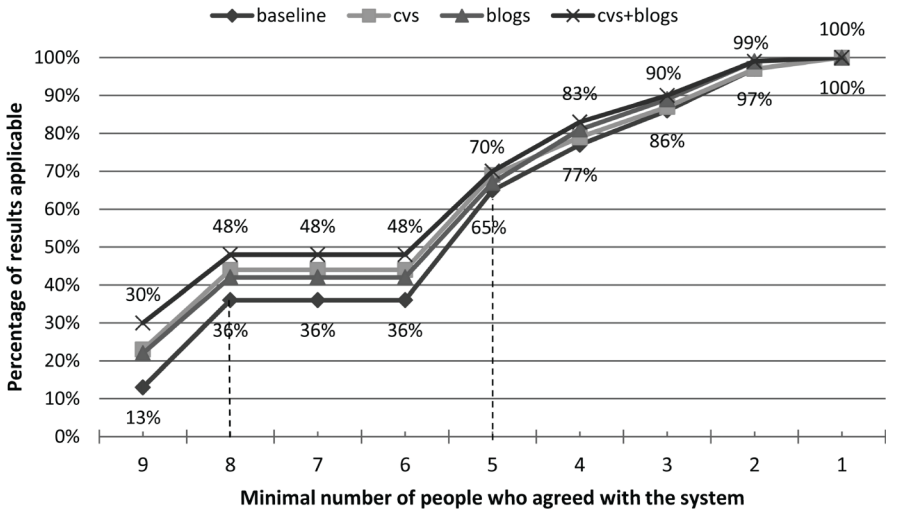
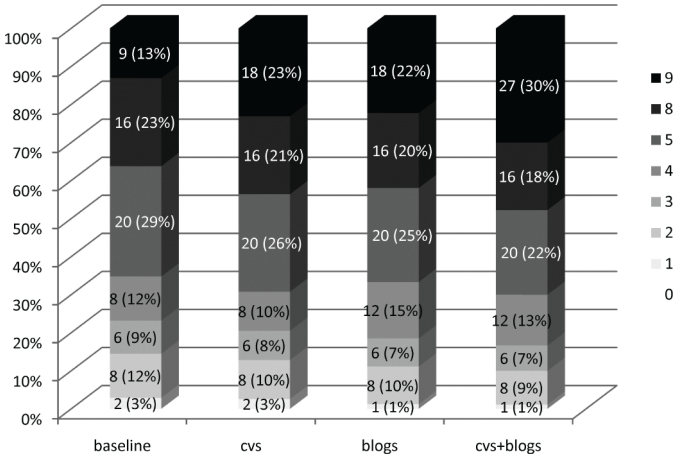


Figure 8. Visualization of the distribution of agreements for all four versions of CAEV procedure (for emotion type determination)



creative human activities, the appropriateness verification should be more difficult to perform in humorous conversations and easier in non-humorous conversations.

In almost all cases the results which changed after the improvements were statistically significant on a 5% level (see Table 11). The only version in which the change of the results was not significant was the baseline method compared to only CVS improvement (P value = 0.1599). Improving the system with blog mining, when compared to both - baseline version of the method and with CVS, were statistically significant (P value = 0.0274) and, what is the most important, the results of the version fully improved were the most significant of all (P value = 0.0119).

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Figure 9. Visualization of percentage of results encapsulated for each condition, from "at least 9" to "at least 1" (for valence determination)

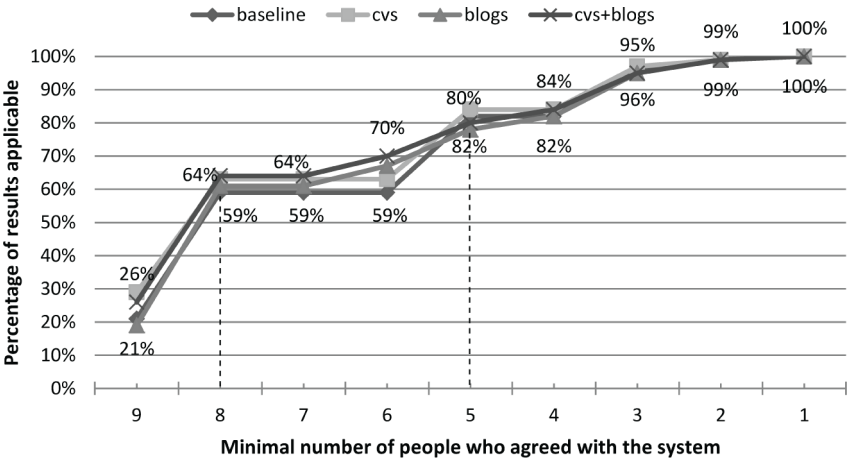
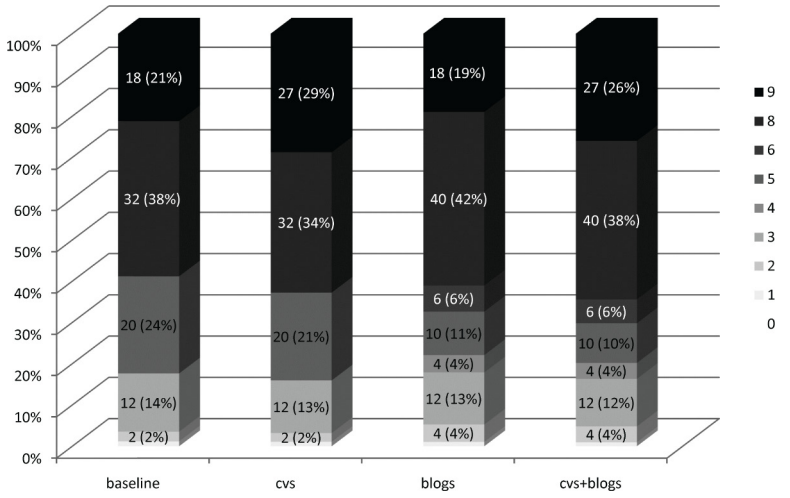


Figure 10. Visualization of the distribution of agreements for all four versions of CAEV procedure (for valence determination)



Although the method for verification of contextual appropriateness of emotion presented here is still not ideal, the increase in results after implementation of different improvements to the intermediary systems (ML-Ask in affect analysis procedure and Web mining) indicate the method is easily improvable. Considering the further enhancements that are already in plans, We are expecting a high improvement of this method in the near future.

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Table 9. Results for evaluation of Contextual Appropriateness of Emotion Verification (CAEV) procedure for Modalin. Description of table contents like in Table 8.

MODALIN								
C) TYPES	Number of people who agreed with the system 8	5	4	3	2	1	0	agr.ratio (kappa)
BASELINE	2	3	1	1	1	1	1	41/100
% of all	39%	37%	10%	7%	5%	2%	0%	41%
% sums	39%	76%	85%	93%	98%	100%	100%	(κ= 0.606666)
CVS	2	3	1	1	1	1	1	41/100
% of all	39%	37%	10%	7%	5%	2%	0%	41%
% sums	39%	76%	85%	93%	98%	100%	100%	(κ= 0.606666)
BLOGS	2	3	2	1	1	0	1	44/100
% of all	36%	34%	18%	7%	5%	0%	0%	44%
% sums	36%	70%	89%	95%	100%	100%	100%	(κ= 0.573333)
CVS+BLOGS	2	3	2	1	1	0	1	44/100
% of all	36%	34%	18%	7%	5%	0%	0%	44%
% sums	36%	70%	89%	95%	100%	100%	100%	(κ= 0.573333)
D) VALENCE	Number of people who agreed with the system 9	8	5	4	3	2	0	agr.ratio (kappa)
BASELINE	1	4	1	0	2	0	2	52/100
% of all	17%	62%	10%	0%	12%	0%	0%	52%
% sums	17%	79%	88%	88%	100%	100%	100%	(κ= 0.688888)
CVS	1	4	1	0	2	0	2	52/100
% of all	17%	62%	10%	0%	12%	0%	0%	52%
% sums	17%	79%	88%	88%	100%	100%	100%	(κ= 0.688888)
BLOGS	1	4	1	0	3	0%	1	55/100
% of all	16%	58%	9%	0%	16%	0%	0%	55%
% sums	16%	75%	84%	84%	100%	100%	100%	(κ= 0.642222)
CVS+BLOGS	1	4	1	0	3	0%	1	55/100
% of all	16%	58%	9%	0%	16%	0%	0%	55%
% sums	16%	75%	84%	84%	100%	100%	100%	(κ= 0.642222)

DISCUSSION

Implications Towards Computational Conscience

Public opinion is a second conscience. -William R. Alger

Conscience is, in most men, an anticipation of the opinions of others. - Henry Taylor

As mentioned above, expressing and understanding emotions is one of the most important cognitive human behaviors present in everyday communication. In particular, Salovey and Mayer (1990) showed that emotions are a vital part of human intelligence, and Schwarz (2000) showed, that emotional states influence the decision making process in humans. If the process of decision making is defined as distinguishing between good and bad, or appropriate and inappropriate, the emotions appear as an influential part of human conscience. The thesis that emotions strongly influence the development of human conscience was proved by Thompson and colleagues (Thompson, Laible, & Ontai, 2003) who showed, that children acquire the conscience by learning the emotional patterns from other people. The significance of the society was pointed out also by Rzepka et al. (2006), who defined the Internet, being a collection of other people's ideas and experiences, as an approximation of general human common sense. Since conscience can be also defined as

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Table 10. Results for evaluation of Contextual Appropriateness of Emotion Verification (CAEV) procedure for Pundalin. Description of table contents like in Table 8.

PUNDALIN										
C) TYPES	Number of people who agreed with the system 9	8	6	5	4	3	2	1	0	agr.ratio (kappa)
BASELINE	1	0	0	1	1	1	3	1	2	28/100
% of all	32%	32%	32%	18%	14%	11%	21%	4%	0%	28%
% sums	32%	32%	32%	50%	64%	75%	96%	100%	100%	(κ= 0.698)
CVS	2	0	0	1	1	1	3	1	1	37/100
% of all	49%	49%	49%	14%	11%	8%	16%	3%	0%	37%
% sums	49%	49%	49%	62%	73%	81%	97%	100%	100%	(κ= 0.678)
BLOGS	2	0	0	1	1	1	3	1	1	37/100
% of all	49%	49%	49%	14%	11%	8%	16%	3%	0%	37%
% sums	49%	49%	49%	62%	73%	81%	97%	100%	100%	(κ= 0.678)
CVS+BLOGS	3	0	0	1	1	1	3	1	0	46/100
% of all	59%	59%	59%	11%	9%	7%	13%	2%	0%	46%
% sums	59%	59%	59%	70%	78%	85%	98%	100%	100%	(κ= 0.658)
D) VALENCE	Number of people who agreed with the system 9	8	6	5	4	3	2	1	0	agr.ratio (kappa)
BASELINE	1	0	0	3	0	2	1	1	2	33/100
% of all	27%	0%	0%	45%	0%	18%	6%	3%	0%	33%
% sums	27%	27%	27%	73%	73%	91%	97%	100%	100%	(κ= 0.664)
CVS	2	0	0	3	0	2	1	1	1	42/100
% of all	43%	0%	0%	36%	0%	14%	5%	2%	0%	42%
% sums	43%	43%	43%	79%	79%	93%	98%	100%	100%	(κ= 0.644)
BLOGS	1	1	1	1	1	1	2	1	1	40/100
% of all	23%	20%	15%	13%	10%	8%	10%	3%	0%	40%
% sums	23%	43%	58%	70%	80%	88%	98%	100%	100%	(κ= 0.644)
CVS+BLOGS	2	1	1	1	1	1	2	1	0	49/100
% of all	37%	16%	12%	10%	8%	6%	8%	2%	0%	49%
% sums	37%	53%	65%	76%	84%	90%	98%	100%	100%	(κ= 0.624)

Table 11. Statistical significance of differences between the results for different versions of the system. The highest significance in bold font.

Versions of the System	Baseline vs CVS	Baseline vs Blogs	CVS vs CVS+Blogs	Baseline vs CVS+Blogs
Statistical Significance (p Value)	0.1599 (p>5%)	0.0274 (p<5%)	0.0274 (p<5%)	**0.0119 (p<5%)**

a part of common sense (moral common sense in particular), this statement can be expanded further to saying that the Web can also be used to determine human conscience. The need for research in this matter, was pointed out *inter alia* by Rzepka et al. (2008), who raised the matter not of creating an artificial human being, as it is popularly ventured in Artificial Intelligence research, but rather an intelligent agent in the form of a toy or a companion, designed to support humans in everyday life. To perform that, the agent needs to be equipped, not only in procedures for recognizing phenomena concerning the user, in which emotions play a great role, but it also needs to be equipped with evaluative procedures distinguishing about whether the phenomena are appropriate or not for a situation the user is in. This is an up to date matter in fields such as Roboethics (Veruggio & Operto, 2006), Human Aspects in Ambient Intelligence (Treur, 2007), and in Artificial Intelligence in general. In our research we

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Table 12. Three examples of the results provided by the emotion appropriateness verification procedure (CAVP) with a separate display of the examples showing the improvement of the procedure after applying CVS

Part of Conversation in Japanese (English Translation)	**ML-Ask Output**		**Web Mining**	**CAEV**
USER: *Konpyūta wa omoshiroi desu ne.* (Computers are so interesting!)	**positive** [joy]		**positive** [joy]	**appro-priate**
SYSTEM: *Sore wa oishii desu ka.* (Is it tasty?) [about instant noodles]	x		x	x
USER: *Oishii kedo, ore wa akita kana.* (Its tasty, but I've grown tired of it.)	**negative** [dislike]		**negative** [dislike]	**appro-priate**
Part of conversation in Japanese (English translation)	**ML-Ask baseline**	**ML-Ask +CVS**	**Web Mining**	**CAEV**
SYSTEM: *Sore wa omoshiroi tte* (It's so funny!) [about conversation]	x	x	x	X
USER: *Sore hodo omoshiroku mo nakatta yo.* (It wasn't that funny.)	positive [joy]	**negative** [dislike]	**negative** [fear], [sad]	**appro-priate**

performed that by verifying emotions expressed by the user with a Web mining technique for gathering an emotional common sense, which could be also defined as an approximated vector of conscience. We understand, that the idea of conscience is far more sophisticated, however, when defined narrowly as the ability to distinguish between what is appropriate and what is inappropriate, our method for verifying contextual appropriateness of emotions could be applied to obtain simplified conscience calculus for machines. We plan to develop further this idea and introduce it as a complementary algorithm for the novel research on discovering morality level in text utterances presented by Rzepka and colleagues (Rzepka, Masui, & Araki, 2009; Komuda, Ptaszynski, Momouchi, Rzepka, & Araki, 2010).

CONCLUSION AND FUTURE WORK

In this paper we presented our work on Contextual Affect Analysis (CAA) with a future goal of its application in conversational agents such as agent companions or artificial tutors. We applied two systems for affect analysis and a Web mining technique to develop a method for enhancement of human-computer interaction in human-agent based dialogs.

The first system for affect analysis we applied was ML-Ask, a system for affect analysis of textual input utterances in Japanese. The system firstly determines whether an utterance is emotive or not and in emotive utterances determines specific emotion types expressed by the user. The second system for affect analysis we applied was CAO, a prototype system for automatic affect analysis of Eastern type emoticons (*kaomoji*). The system was created using a database of over ten thousand of unique emoticons automatically extracted from the Internet and expanded to over three million emoticon combinations.

The above two systems are used in an affect analysis procedure applied in a method for estimating contextual appropriateness of emotions. The method is composed of two parts, a language based affect analysis procedure utilizing the two affect analysis systems (used as an emotion detector), and a Web mining technique for extracting from the Internet lists of emotional associations considered as a generalized emotive common sense (used as an

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emotion verifier). We checked the performance of four versions of the method. The affect analysis procedure is compared with and without Contextual Valence Shifters. As for the Web mining technique, two versions are compared: one, using all of the Internet resources and the second one improved by restricting the search scope to the contents of blog documents. The improvements positively influenced the results and were statistically significant. We also observed that emotion appropriateness was more difficult to determine in conversations containing humorous responses (puns).

The method provides the conversational agent with computable means to determine whether emotions expressed by a user are appropriate for the context they appear in. A conversational agent equipped with this method could be provided with hints about what communication strategy would be the most desirable at a certain moment. For example, a conversational agent could choose to either sympathize with the user or take precautions and help the user manage his/her emotions.

By proposing computational means for verification of contextual appropriateness of emotional states in conversation, this research defines a new set of goals for Affective Computing. In particular we proposed a set of tasks to be addressed with regards to broadening the scope of Affect Analysis. The tasks, which we include under a general term Computational Fronesis (CF), include: determination of emotion subject, object, verification of contextual appropriateness of emotional expression and its degree, and modeling of behavior appropriate for the situation or context.

We showed that computing emotions in context is a feasible task. Although the methods are still not ideal and its components (ML-Ask, CAO, the Web mining technique) need further improvement, there have been seen a significant improvement, e.g., by restricting Web mining to the contents of *Yahoo!Japan-Blogs*. As for the future work, we plan to focus on deepening the understanding of emotions by bootstrapping the context phrases. For example, in a sentence "I'm so depressed since my girlfriend left me..." the context phrase would be "my girlfriend left me". The Web mining procedure provides for such phrases a list of appropriate emotions. However, using similar Web mining procedure we plan to go further and find out the reason for an emotion object to happen. For example, to find out "why girls leave their boyfriends?". An answer for this question, found in the Internet, could be, e.g., "because boys are not sporty enough", or "because boys have no money". Next asked question could be, e.g., "why boys have no money?", etc. Sufficient accuracy in such bootstrapping method would provide a deeper knowledge about the causality of experiences. When applied in artificial tutor, or a companion agent in general, this would help providing hints about predictable undesirable consequences of user activities.

ACKNOWLEDGMENT

This research was supported by (JSPS) KAKENHI Grant-in-Aid for JSPS Fellows (Project Number: 22-00358).

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ENDNOTES

1 We understand that the notion of fronesis is far more sophisticated than the one explained here, but for practical reasons we generalize the term and limit its scope to emotion-related processing.
2 Separation by "/" represents two possible readings of the character.
3 In particular Japanese emoticons called *kao-moji*.
4 blogs.yahoo.co.jp
5 Abbreviation from Contextual Appropriateness of Emotions Verification
6 http://search.goo.ne.jp/

Michal Ptaszynski was born in Wroclaw, Poland in 1981. He received the MA degree from the University of Adam Mickiewicz, Poznan, Poland, in 2006, and PhD in Information Science and Technology from Hokkaido University, Japan in 2011. At the moment he is a JSPS Post-doctoral Research Fellow at the High-Tech research Center, Hokkai-Gakuen University, Japan. His research interests include natural language processing, dialogue processing, affect analysis, sentiment analysis, HCI, and information retrieval. He is a member of the ACL, the AAAI, the IEEE, the HUMAINE, the AAR, the SOFT, the JSAI, and the ANLP.

Pawel Dybala was born in Ostrow Wielkopolski, Poland in 1981. He received his MA in Japanese Studies from the Jagiellonian University in Krakow, Poland in 2006, and PhD in Information Science and Technology from Hokkaido University, Japan in 2011. Currently he is a director and general project manager at Kotoken Language Laboratory in Krakow. His research interests include natural language processing, dialogue processing, humor processing, HCI, and information retrieval.

Michal Mazur was born in Krakow, Poland in 1981. He received his MA in Language Education from the Pedagogical University in Krakow, Poland in 2005. He is studying towards his PhD at Graduate School of Information Science and Technology, Hokkaido University. His research interests include e-learning, artificial tutoring systems, natural language processing, dialogue processing, HCI, and information retrieval.

Rafal Rzepka received the MA degree from the University of Adam Mickiewicz, Poznan, Poland, in 1999, and the PhD degree from Hokkaido University, Japan, in 2004. Currently, he is an assistant professor in the Graduate School of Information Science and Technology at Hokkaido University. His research interests include natural language processing, Web mining, common sense retrieval, dialogue processing, language acquisition, affect analysis, and sentiment analysis. He is a member of the AAAI, the ACL, the JSAI, the IPSJ, the IEICE, the JCSS, and the ANLP

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Kenji Araki received the BE, ME, and PhD degrees in electronics engineering from Hokkaido University, Sapporo, Japan, in 1982, 1985, and 1988, respectively. In April 1988, he joined Hokkai-Gakuen University, Sapporo, Japan, where he was a professor. He joined Hokkaido University in 1998 as an associate professor in the Division of Electronics and Information Engineering and became a professor in 2002. Presently, he is a professor in the Division of Media and Network Technologies at Hokkaido University. His research interests include natural language processing, spoken dialogue processing, machine translation, and language acquisition. He is a member of the AAAI, the IEEE, the JSAI, the IPSJ, the IEICE, and the JCSS.

Yoshio Momouchi was born in 1942 in Hokkaido, Japan. He obtained a master's degree and a doctorate in engineering from Hokkaido University. He was a member of the Division of Information Engineering in Graduate School at Hokkaido University from 1973 to 1988. Since 1988 he has been a professor in the faculty of engineering at Hokkai-Gakuen University. He fulfilled duties of Dean of Graduate School of Engineering at Hokkai-Gakuen University in years 2005-2008. He specializes in intelligent information processing, computational linguistics and machine translation. He is a member of the IPSJ, the ANLP, the ACL, the MLSJ, the JCSS, and the JSAI.

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